Mika Vohl

mikavohl@gmail.com | linkedin.com/in/MikaVohl | github.com/MikaVohl | mikavohl.ca

Education

McGill University — September 2023 – May 2027

Bachelor of Science, Joint Honours in Computer Science and Physics — *Montreal, Canada*

- **Cumulative GPA:** 3.93/4.0 - Dean's Honour List 2023/2024
- **President - Hack4Impact:** Led a 45+ member student collective to build full-stack applications for nonprofits
- **Developer - McHacks:** Built and maintained software to operate McGill's largest hackathon (500+ users)

Technical Skills

Languages: Python, Go, C++, Java, C, JavaScript, TypeScript, SQL, Bash, HTML/CSS, PHP
Frameworks & Libraries: PyTorch, CUDA, ROCm, Flask, React, Tensorflow/Keras, Matplotlib, Pandas, scikit-learn
Developer Tools: Git, Docker, Kubernetes, Linux, Github Actions, AWS, Hugging Face, PostgreSQL, llama.cpp
Certifications: Stanford Machine Learning Specialization, AMD HIP GPU Programming

Experience

Machine Learning Libraries Intern — May 2026 – Present

AMD — *Calgary, Canada*

- Improved GEMM kernel selection performance by 4.3% by building an ML-based optimization pipeline that replaces hard-coded heuristics with data-driven parameters, reaching $>$ 6% gains with workload-specific heuristics
- Automated collection and analysis of GPU performance counters and execution traces during GEMM tuning, revealing compute, memory, occupancy, and execution bottlenecks at the kernel level
- Tuned GEMM kernels for OpenAI, Meta, and Mistral AI models on AMD GPUs, reducing latency by up to 10%
- Shipped features and improvements to AMD's open-source GPU libraries serving thousands of enterprise users

Researcher — September 2025 – July 2026

McGill AI Society — *Montreal, Canada*

- Researching deep learning on missing clinical data; supervised by Mila grad student; accepted to AI4PH poster fair
- Benchmarked TabPFN and ConTextTab models against MLP/XGBoost on predicting mortality under missingness

Software Engineer Intern (Machine Learning) — May 2025 – Aug. 2025

streamwise.gg — *Montreal, Canada*

- Fine-tuned an open-source Transducer model for ASR in livestreams, cutting transcription error from 26% to 10%
- Built a low-latency audio/video pipeline using Go and FFmpeg, handling transcoding, buffering, multiqueueing
- Containerized and orchestrated services with Docker & Kubernetes, managing networking, scaling, and lifecycles

Software Engineer Intern — May 2024 – Sept. 2024

myBonum — *Toronto, Canada*

- Built an API for interfacing with an SQL database, enabling core backend functionality for a healthcare startup
- Integrated Stripe billing and subscriptions directly into user-facing dashboard and backend infrastructure

Software Developer Intern — June 2023 – Sept. 2023

CAIL — *Toronto, Canada*

- Shipped features and optimizations for an HP NonStop terminal emulator, including Linux/Android support and an 80% throughput improvement via on-demand redraws, optimized sorting algorithms, and buffered I/O.

Projects

Stitch - 1st place winner at McGill AI Society Hacks 25-26 - GitHub | *Typescript, React, Python* — November 2025

- Built a drag-and-drop platform to design, train, and test neural networks for handwritten digit recognition
- Provided users end-to-end control of real PyTorch models by dynamically compiling and executing custom model instances from user specifications, streaming live results back to the user via Server-Sent Events

Predicting Chaos with Convolutional Neural Nets - Try it out! | *Python, PyTorch, Typescript* — May 2025

- Trained a CNN to predict the 5th next state of Conway's Game of Life, a chaotic system, with 98% accuracy
- Built a custom Game of Life engine to generate 50,000 synthetic board states for training a convolutional network

PyTorch-Style Tensor Autograd Engine - GitHub | *Python, Numpy* — June 2025

- Built an autograd engine for propagating gradients across ML computations, supporting tensor-wise operations
- Significantly accelerated neural network computations using Numpy's BLAS vectorized operations